

Module 3 Lesson 3

Linear Functions and Their Graphs (7.2)

Objectives

In this lesson, we will learn how to

- use intercepts to graph a linear equation;
- calculate slope;
- use the slope and y -intercept to graph a line;
- graph horizontal and vertical lines;
- interpret slope as a rate of change;
- use slope and y -intercept to model data.

1. Linear Equations in Two Variables

All equations of the form

$$Ax + By = C$$

are straight lines when graphed, as long as A and B are not both zero. These equations are called linear equations in two variables.

Linear Equation in Two Variables

A linear equation in two variables can be written as

$$Ax + By = C,$$

where A , B , and C are real numbers, and A and B are not both zero.

2. Graphing Using Intercepts

Intercepts are points where a graph crosses an axis.

Locating Intercepts

- To locate the **x-intercept**, set $y = 0$ and solve for x .
- To locate the **y-intercept**, set $x = 0$ and solve for y .

The x -intercept has the form

$$(a, 0).$$

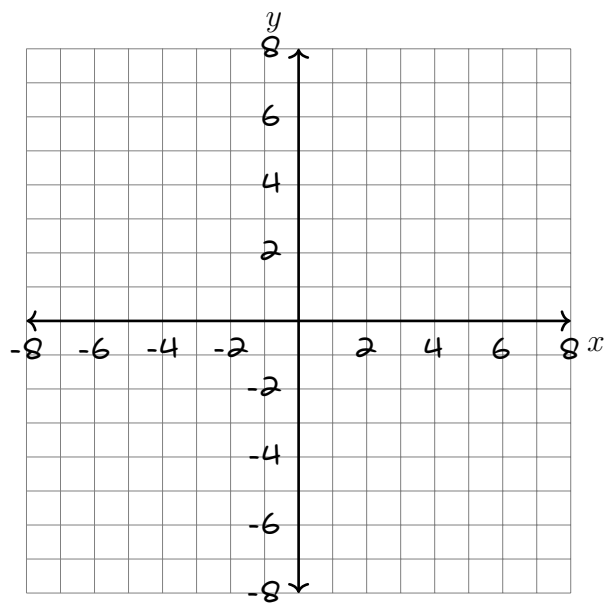
The y -intercept has the form

$$(0, b).$$

3. Using Intercepts to Graph a Linear Equation

Example 1: Graph the equation using intercepts:

$$3x + 2y = 6.$$



Solution

To find the x -intercept, set $y = 0$:

$$3x + 2(0) = 6.$$

$$3x = 6.$$

$$x = 2.$$

So the x -intercept is

$$(2, 0).$$

To find the y -intercept, set $x = 0$:

$$3(0) + 2y = 6.$$

$$2y = 6.$$

$$y = 3.$$

So the y -intercept is

$$(0, 3).$$

Plot $(2, 0)$ and $(0, 3)$, then draw a line through the two points.

To check, substitute $x = 1$:

$$3(1) + 2y = 6.$$

$$3 + 2y = 6.$$

$$2y = 3.$$

$$y = \frac{3}{2} = 1.5.$$

So another point on the line is

$$(1, 1.5).$$

4. Slope

The **slope** of a line measures its steepness.

Slope is the ratio of vertical change to horizontal change.

Slope Formula

The slope of the line through two distinct points

$$(x_1, y_1)$$

and

$$(x_2, y_2)$$

is

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

where

$$x_2 - x_1 \neq 0.$$

Slope is often described as

$$m = \frac{\text{rise}}{\text{run}}.$$

5. Finding Slope from Two Points

Example 2: Find the slope of the line passing through the points

$$(-3, -1)$$

and

$$(-2, 4).$$

Solution

Let

$$(x_1, y_1) = (-3, -1)$$

and

$$(x_2, y_2) = (-2, 4).$$

Use the slope formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Substitute:

$$m = \frac{4 - (-1)}{-2 - (-3)}.$$

Simplify:

$$m = \frac{5}{1} = 5.$$

The slope is

$$5.$$

6. Slope-Intercept Form

The slope-intercept form of the equation of a nonvertical line is

$$y = mx + b.$$

Slope-Intercept Form

The equation

$$y = mx + b$$

has slope m and y -intercept b .

The y -intercept is the point

$$(0, b).$$

7. Graphing Using Slope and y -Intercept

Graphing $y = mx + b$

To graph a line using slope and y -intercept:

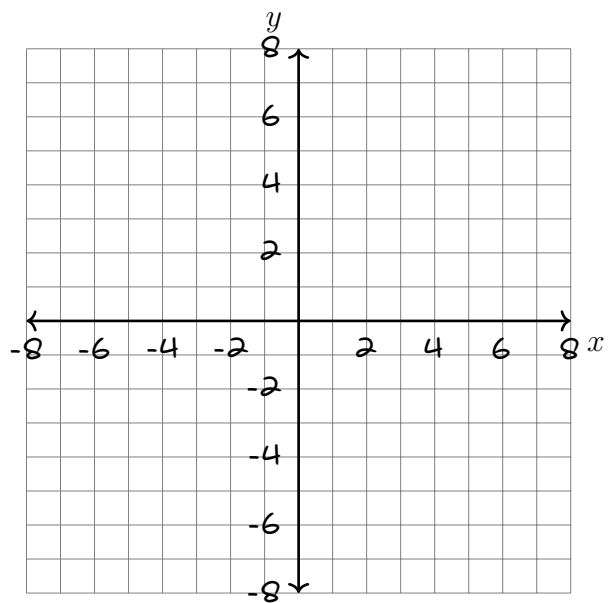
1. Plot the y -intercept $(0, b)$.
2. Write the slope m as a fraction:

$$m = \frac{\text{rise}}{\text{run}}.$$

3. From the y -intercept, use the rise and run to find a second point.
4. Draw a line through the two points.

Example 3: Graph the linear function using the slope and y -intercept:

$$y = \frac{2}{3}x + 2.$$



Solution

The equation is in slope-intercept form:

$$y = mx + b.$$

So,

$$m = \frac{2}{3}$$

and

$$b = 2.$$

The y -intercept is

$$(0, 2).$$

The slope is

$$\frac{\text{rise}}{\text{run}} = \frac{2}{3}.$$

Starting at $(0, 2)$, move up 2 units and right 3 units.

This gives the second point

$$(3, 4).$$

Draw a line through

$$(0, 2)$$

and

$$(3, 4).$$

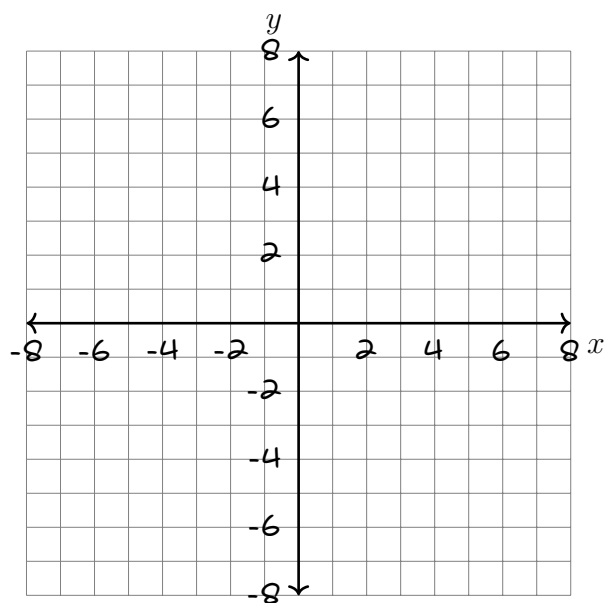
8. Rewriting in Slope-Intercept Form

Sometimes an equation is not given in slope-intercept form.

In that case, solve the equation for y .

Example 4: Graph the linear function using the slope and y -intercept:

$$2x + 5y = 0.$$



Solution

Solve for y :

$$2x + 5y = 0.$$

Subtract $2x$ from both sides:

$$5y = -2x.$$

Divide by 5:

$$y = -\frac{2}{5}x.$$

So the slope is

$$m = -\frac{2}{5}$$

and the y -intercept is

$$(0, 0).$$

Starting at $(0, 0)$, move down 2 units and right 5 units.
This gives the second point

$$(5, -2).$$

Draw a line through

$$(0, 0)$$

and

$$(5, -2).$$

9. Horizontal Lines

A horizontal line has an equation of the form

$$y = b.$$

All points on the line have the same y -coordinate.

Horizontal Line

The graph of

$$y = b$$

or

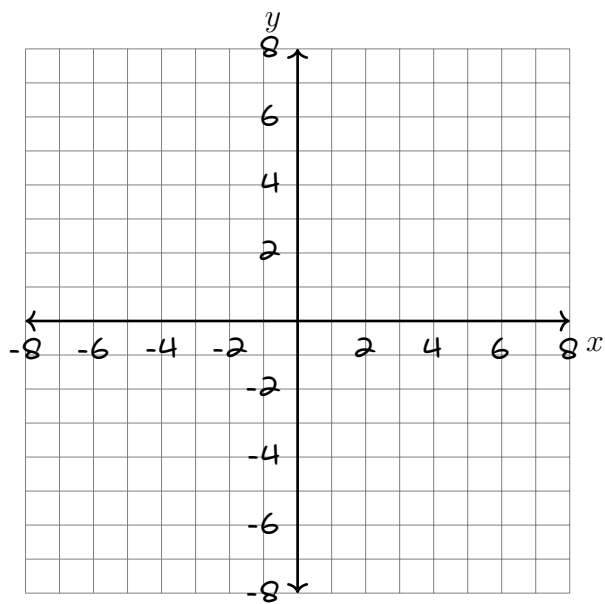
$$f(x) = b$$

is a horizontal line.

The y -intercept is b .

Example 5: Graph the horizontal line:

$$y = -4.$$



Solution

All ordered pairs on the graph have y -coordinate -4 .
Examples of points on the line are

$$(-2, -4), \quad (0, -4), \quad (3, -4).$$

Plot these points and draw a horizontal line through them.

10. Vertical Lines

A vertical line has an equation of the form

$$x = a.$$

All points on the line have the same x -coordinate.

Vertical Line

The graph of

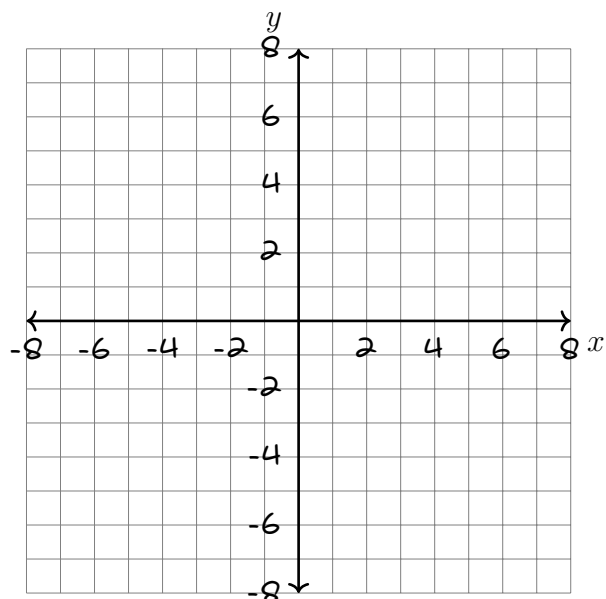
$$x = a$$

is a vertical line.

The x -intercept is a .

Example 6: Graph the vertical line:

$$x = 2.$$



Solution

All ordered pairs on the graph have x -coordinate 2.

Examples of points on the line are

$$(2, -2), \quad (2, 0), \quad (2, 3).$$

Plot these points and draw a vertical line through them.

11. Horizontal and Vertical Lines

Summary of Horizontal and Vertical Lines

Equation	Graph	Function?
$y = b$	Horizontal line	Yes
$x = a$	Vertical line	No

No vertical line represents a function of x .

All nonvertical lines are graphs of functions.

12. Slope as a Rate of Change

Slope is defined as a ratio of a change in y to a corresponding change in x .

$$m = \frac{\text{change in } y}{\text{change in } x}.$$

In applied situations, slope can be interpreted as a rate of change.

Slope as Rate of Change

If x represents time and y represents a quantity that changes over time, then the slope represents how much y changes for each one-unit increase in x .

Example 7: In 1980, about 21% of college freshmen identified as liberal.

In 2016, about 36% identified as liberal.

Find the slope of the line segment connecting the points

(1980, 21)

and

(2016, 36).

Round to one decimal place. Describe what the slope represents.

Solution

Use the slope formula:

$$m = \frac{36 - 21}{2016 - 1980}.$$

$$m = \frac{15}{36}.$$

$$m \approx 0.4.$$

The slope means that from 1980 through 2016, the percentage of liberal college freshmen increased by approximately

$$0.4\%$$

per year.

13. Modeling Data with Linear Functions

Linear functions are useful for modeling data that fall on or near a line.

A linear model often has the form

$$y = mx + b.$$

Linear Model

In a linear model

$$y = mx + b,$$

- m represents the rate of change;
- b represents the initial value or starting amount.

Example 8: Suppose a data set is modeled by

$$y = 3x + 12.$$

- What is the slope?
- What is the y -intercept?
- What does the model predict when $x = 5$?

Solution

The equation is in the form

$$y = mx + b.$$

So the slope is

$$m = 3.$$

The y -intercept is

$$b = 12.$$

When $x = 5$,

$$y = 3(5) + 12.$$

$$y = 15 + 12 = 27.$$

The model predicts

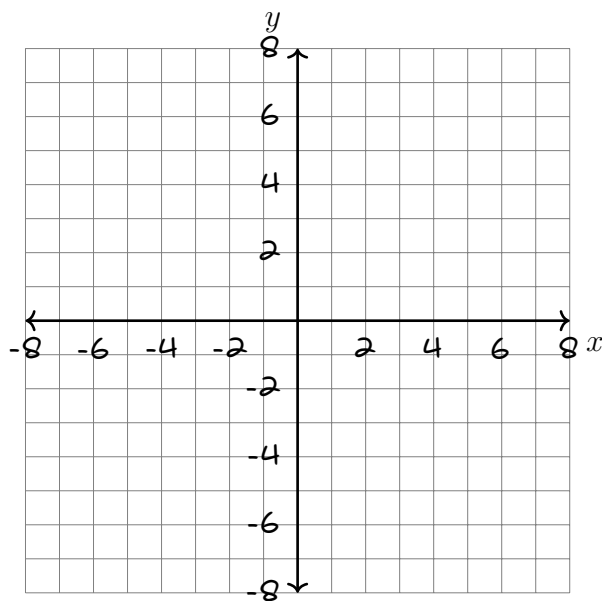
$$y = 27.$$

14. More Practice

Example 9: Find the x -intercept and y -intercept of

$$4x - 2y = 8.$$

Then graph the line.



Example 10: Find the slope of the line passing through

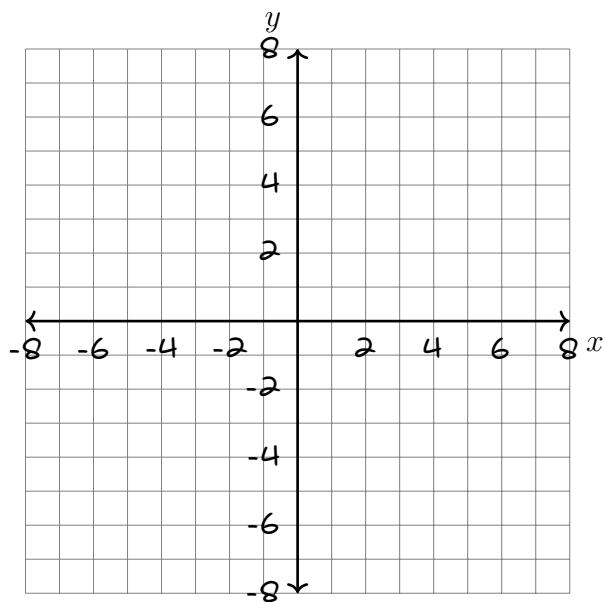
$$(1, 5)$$

and

$$(4, -1).$$

Example 11: Graph the line using slope and y -intercept:

$$y = -\frac{3}{4}x + 1.$$



Summary

Key Ideas

- Linear equations in two variables graph as straight lines.
- To find the x -intercept, set $y = 0$.
- To find the y -intercept, set $x = 0$.
- Slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

- Slope-intercept form is

$$y = mx + b.$$

- The graph of $y = b$ is a horizontal line.
- The graph of $x = a$ is a vertical line.
- Vertical lines are not functions of x .
- Slope can be interpreted as a rate of change.
- Linear models can be used to describe data that are approximately linear.